How to Model Dynamical Systems

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Final Presentation Forschungspraxis

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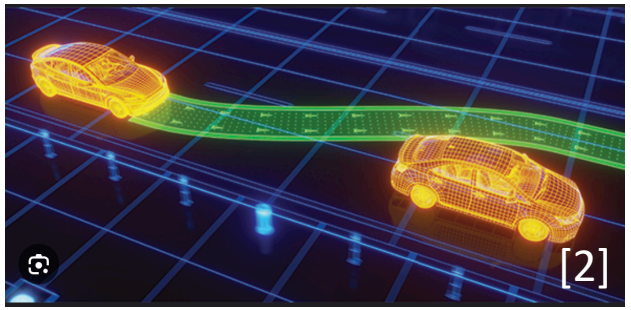
Motivation

Why is time series prediction important?

Business & Finance

Autonomous Systems



Weather & Climate Science

Better prediction -> better decision

[1] : https://www.finanzen.net/index/msci-world
[2]: https://read.nxtbook.com/
[3]: https://www.alamy.com/

Problem Formulation

When to use which method ?

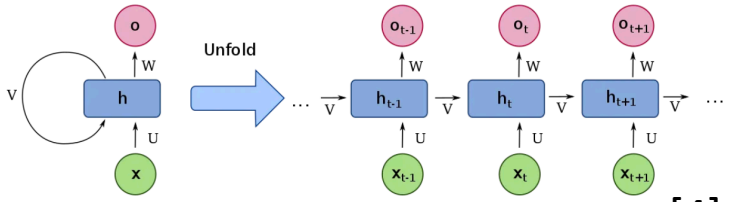
- Too many model options
- Lack of systematic classification

Goal :

- Compare models systematically
- Enable better model selection

Model classes

RNN- state based



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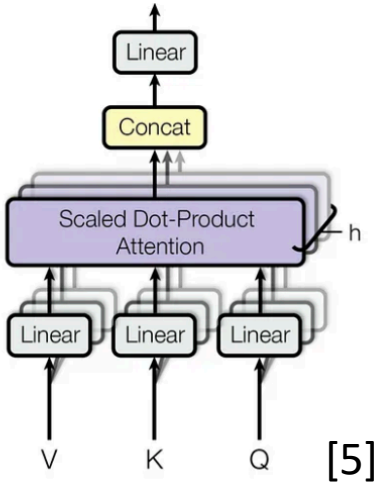
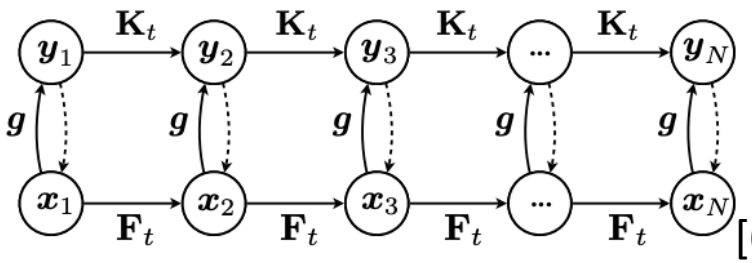
Transformer- attention based



[5]

Koopman- operator based



[6]

[4]: https://medium.com/@hadarsharvit/what-is-a-recurrent-neural-network-elman-net-part-1-d9a70d8e0b95

[5]: Attention is all you need, 2017

[6]: Modern Koopman theory for dynamical systems, Steven L. Brunton

Datasets

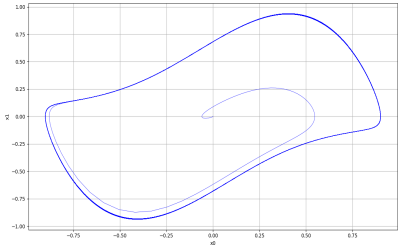
Oscillatory

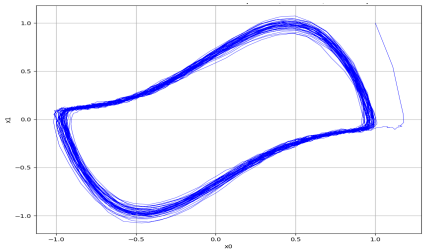
Noisy

Van der Pol oscillator

Van der Pol oscillator with white noise

Limit cycle

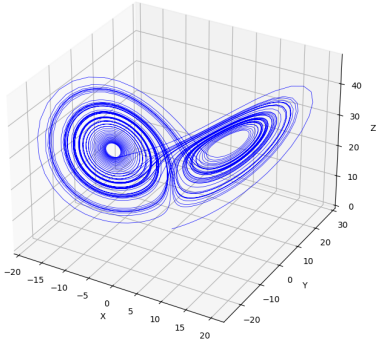


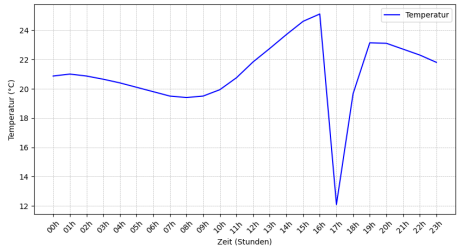


Lorenz System

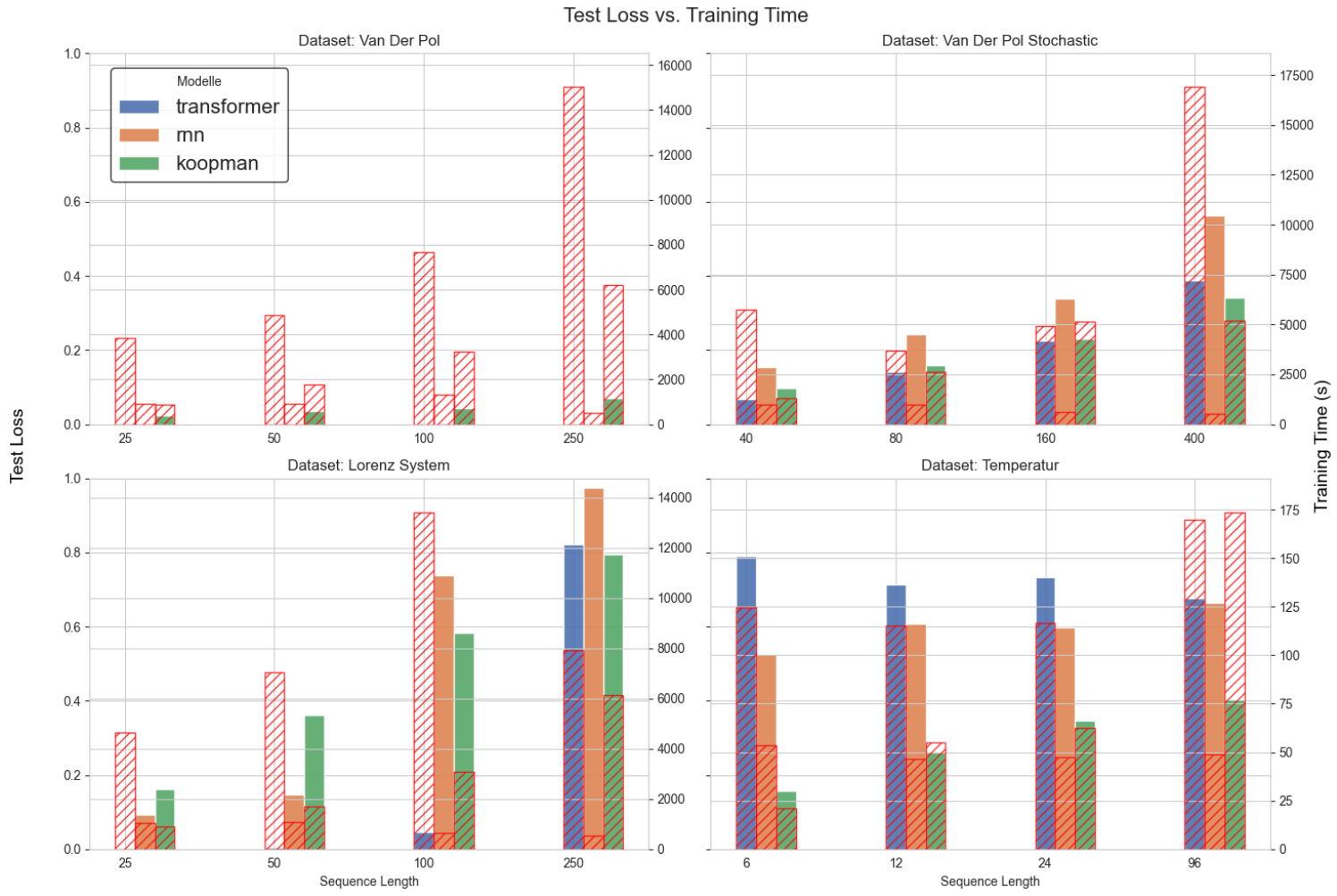
Room temperature

Deterministic





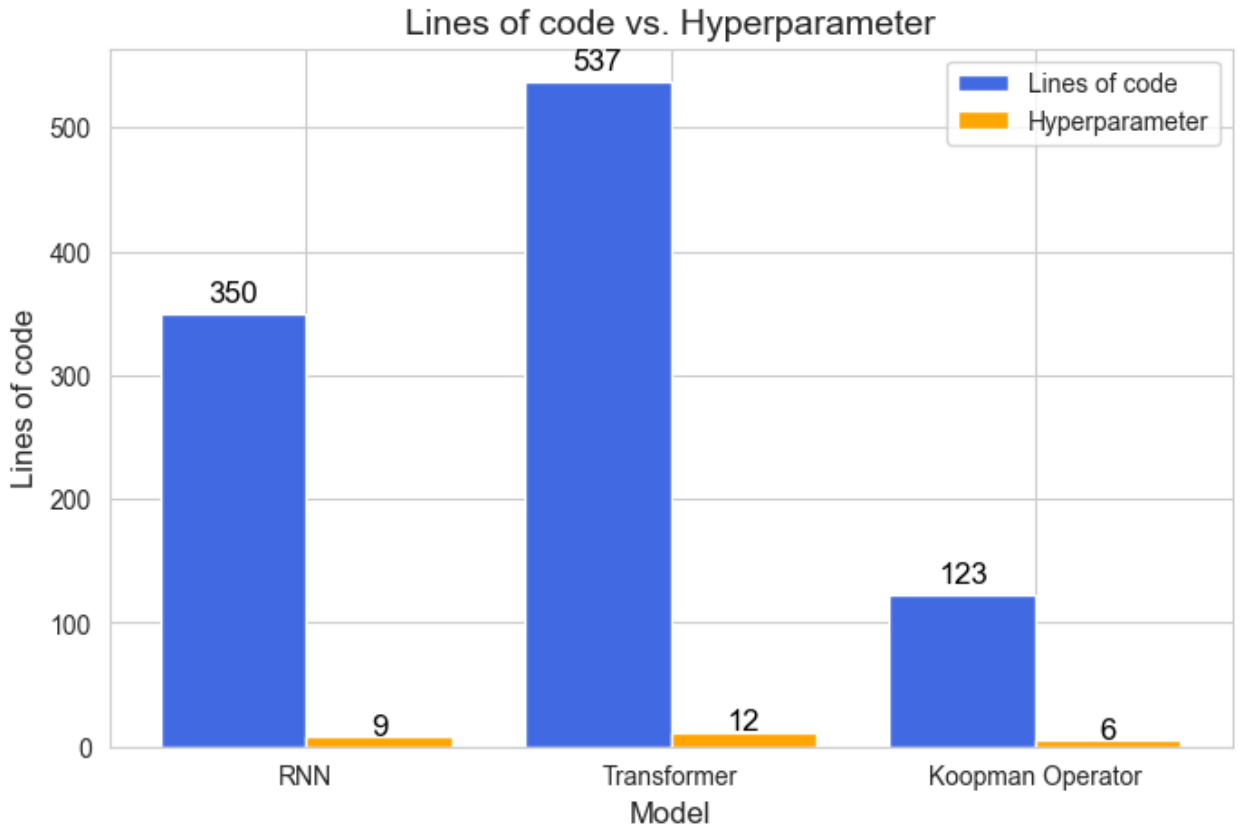
Results 1



- RNN: fast & efficient
- Transformer: accurate but expensive
- Koopman: system-dependent

Results 2

- Transformer: complex
- RNN: moderate effort
- Koopman: simple and easy to optimize



Summary

Systematic comparison:

- Model Architectures: Sequential, Attention, Operator
- Dynamical Systems: limit cycle, strange attractor, noisy, deterministic

Model selection:

- RNN – efficient, good for deterministic systems
- Transformer – complex architecture, high computational costs, good for noisy systems
- Koopman – fast, simple implementation, good for systems with stable structures, competitive for very long sequences